Computer Science & IT

COMPUTER NETWORK
(CN)

Application Layer Protocols

Lecture No. 02



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Recap of Previous Lecture



Topic HTTP

Topic

Topic

Topics to be Covered



- Topic — HTTP
- Topic — NAT ✓
- Topic

HTTP request message :-

The first line of request message is called request line. There are 3 fields in this line seperated by a space. The fields are

1. method
2. URL
3. version of HTTP → 1.1

method	Action
GET	Request a document from the server
Gate HEAD	Requests information about the document but not the document itself
PUT	Sends a document from client to server
POST	Sends information from client to server
TRACE	Echoes the incoming request, client asks the server to echo back the request to check whether the server is getting the requests.

DELETE Removes a webpage

OPTIONS allows the client to ask about the properties of a webpage

HTTP Response message :-

A Response message Consists of a status line, header lines, a blank line and sometimes a body.

The first line in the Response message is called the status line. There are fields in this line seperated by spaces:

1) The first field is version of HTTP → 1.1

2) The Status field defines the status of the request
It is a 3 digit code
The codes in 100 range are only informational
The codes in 200 range indicate a Successfull request
The Codes in 300 range redirects the client to another URL
Codes in 400 range indicate an error at the client side
Codes in 500 range indicate an error at the server side.

101, 102, 103

201, 202, 203

301, 302, 303

401

3) The status phase that explains the status code in text form

→ Body :- The body contain the document to be sent from server to client. The body is present unless the response is an end message

Conditional request :-

A client can add a condition in its request. In this case server will send the requested webpage only if the condition is met & inform the client otherwise. Common condition imposed by a client is the time and date of the web page modified. The client can send the header line if_modified_since with the request to tell the server that it needs a web page only if it is modified after a certain point in time.

Cookies :-

The www was diginally designed as stateless entity A client sends a request, the server responds and their relationship is over.

To remember some information about the clients, a server may need a mechanism in the following situation

1) Websites like amazon that allows users to browse through the store, select the wanted items, put items in a cart and pay at the end with a Credit Card
2) Some websites need to allow access to registed client only
3) Some websites are portals, the user selects the webpages he wants to see
4) Some websites like advertising agencies.

Creation and Storing Cookies :-

The Creation and storing of cookies depend on the implementation; how ever the principle is same

1) when a server receives a request from a client, it stores information about the client in a file of a string. This information may include the domain name of the client, the contents of the cookie (information the server has gathered about the client such as name, registration number and so on), a timestamp and other information depending on the implementation

2) The server includes the cookie in the response that it sends to the client

3) whenever the client receives the response, the browser stores the cookie in the cookie directory, which is sorted by server domain name.

Using Cookies :-

When a client sends a request to a server, the browser looks into cookie directory to see if it can find a cookie sent by the server. If found, the cookie is included in the request. When the server receives the request, it will know that this is an old client, not a new one.

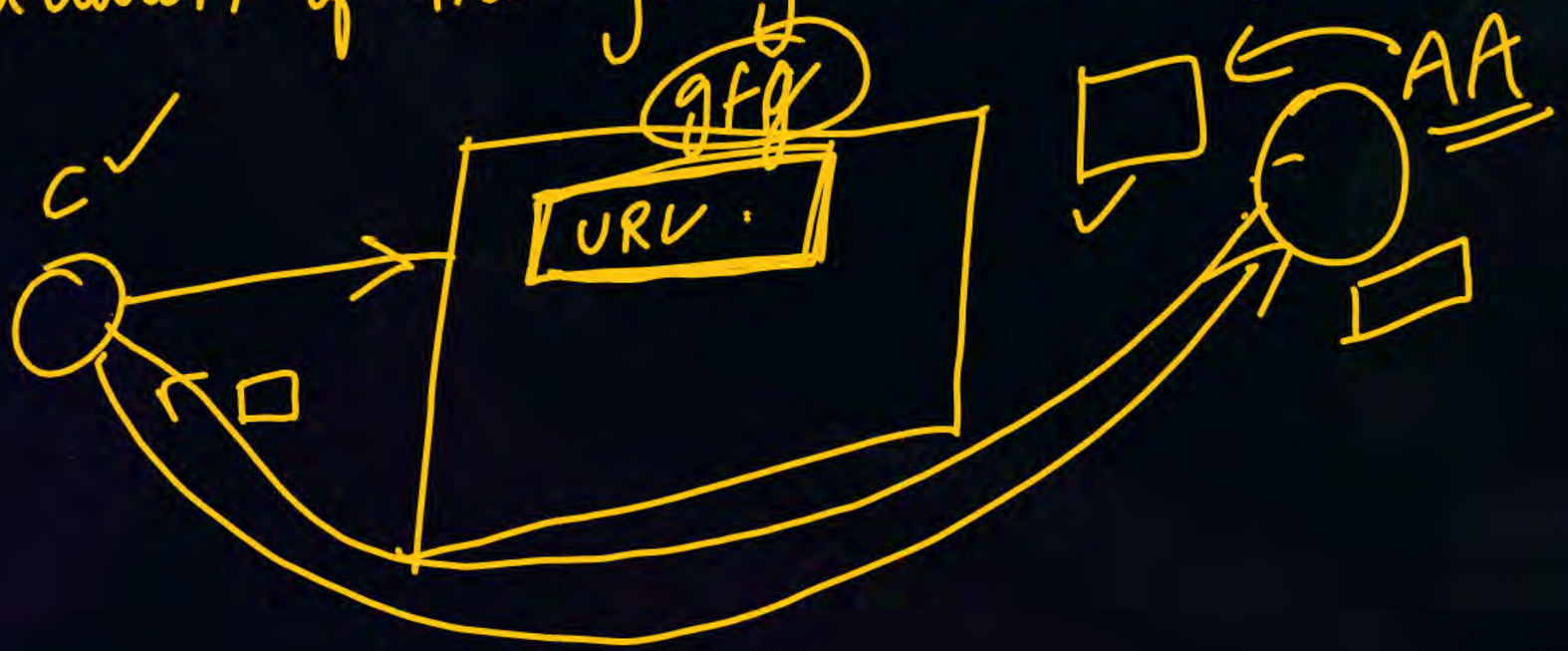
How different websites use cookies :-

1) An electronic store can use a cookie for its client shoppers. when a client selects an item and adds to cart a cookie containing information about the item such as its number and price is sent to the browser. If the client selects the second item, the cookie is updated with new selection information and so on. when the client finishes the shopping and wants to checkout, the last cookie is retrieved and total amount is calculated.

→ The sites that restricts access to registered clients only sends a cookie to the client when client registers for the first time. For any repeated access, only those clients that send the appropriate cookie are allowed

→ A web portal uses the cookie in a similar way when a user selects a page, the cookies is made and sent. If the site is accessed again, the cookie is sent to the server to show what the client is looking for

→ A cookie is also used by advertising agencies
An advertising agency, can place banner ads
on some main websites that is often visited by users.
The advertising agencies supply only a URL, that gives
the address of the agency instead of banner itself.

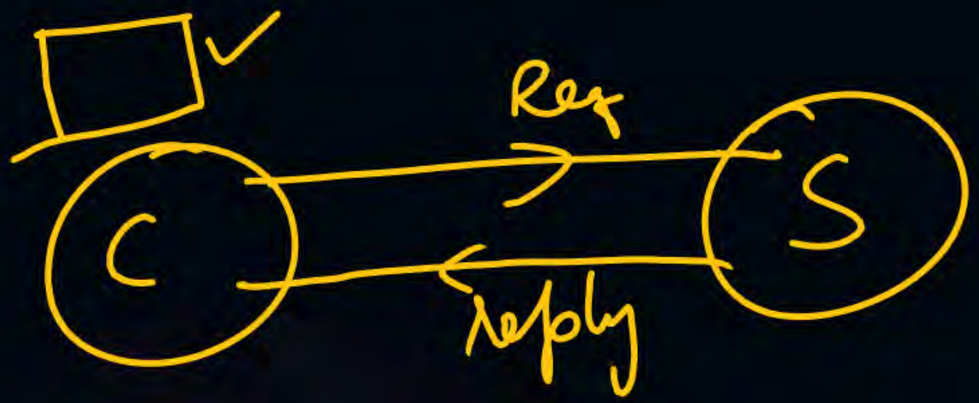


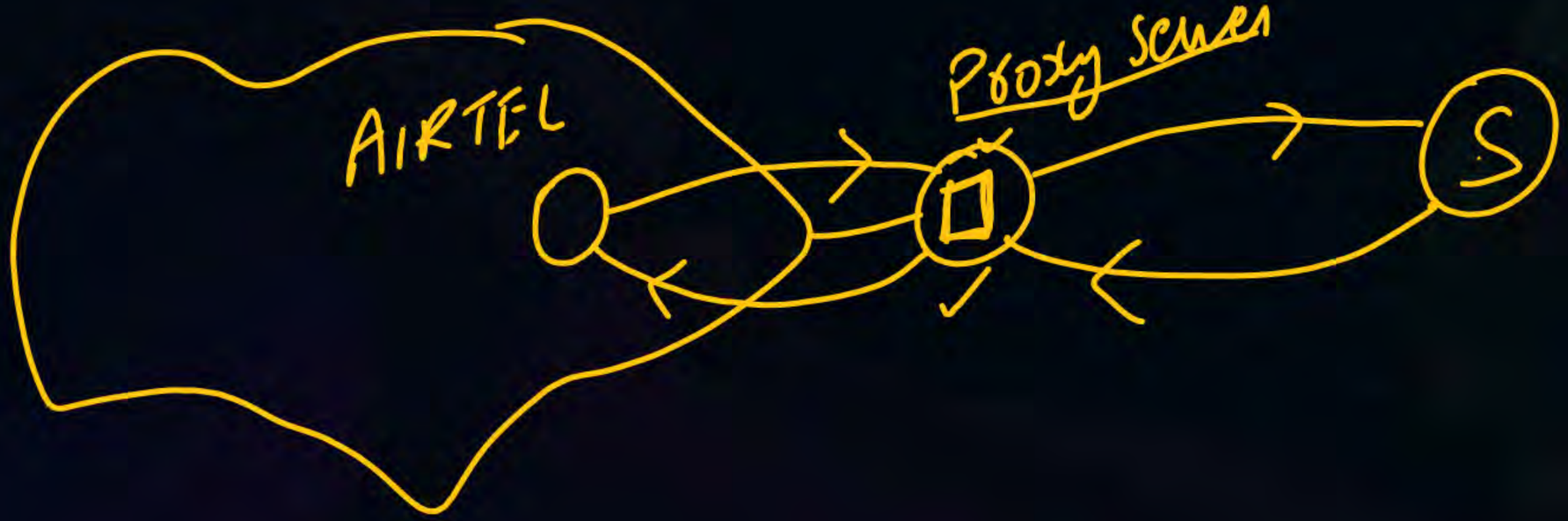
When a user visits the main site, and clicks on the on the icon of the corporation, a request is send to the Advertising agency. The agency sends a requested banner but it also includes a cookie with the ID of the user Any further use of banner ads, adds to the database of the profile of the user.



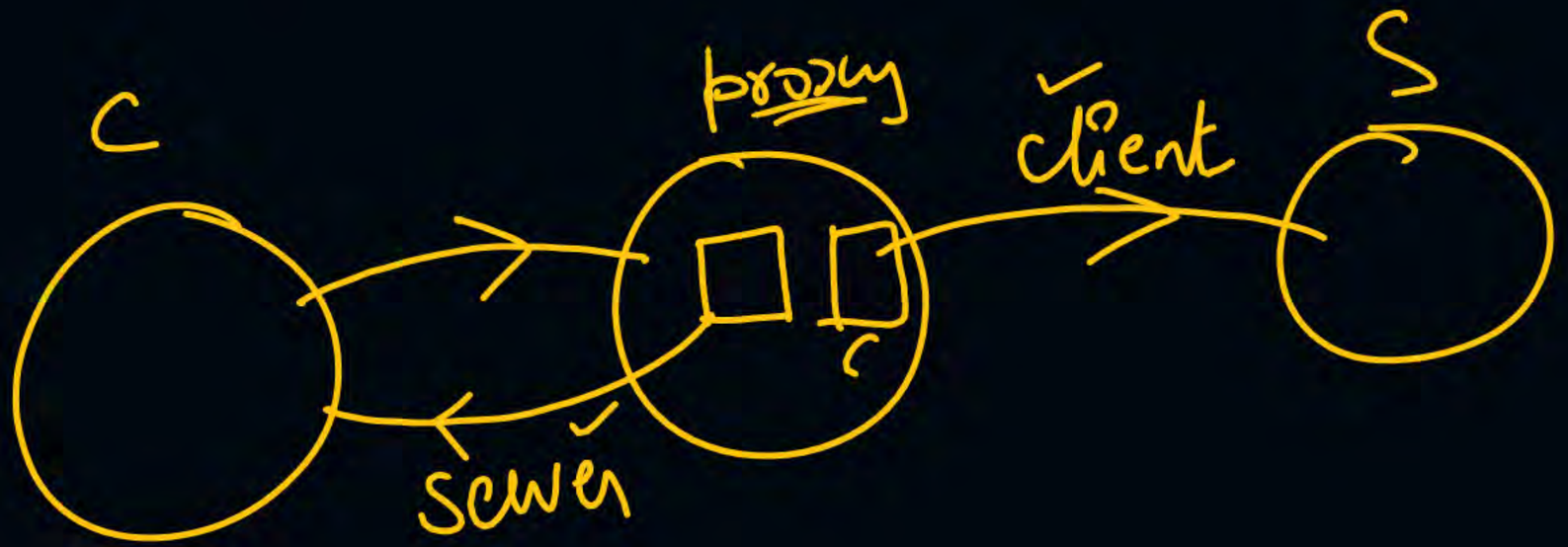
The advertising agency has compiled the interests of the user and can sell this information to other parties. This use of cookie has made them very controversial. Hopefully some regulations will be devised to preserve the privacy of the users

Web caching and proxy servers :-





HTTP Supports proxy servers. A proxy server is a computer that keeps copies of responses to recent requests. The HTTP client send a request to proxy servers. The proxy server checks its cache, if the response is not stored in the cache, the proxy server sends the request to the corresponding server. The proxy server reduces the load on the original server, decreases the traffic and improves latency. Not that proxy can act as both client and server


C
proxy
client
S
server

THANK - YOU